

# Image Retrieval with Contrastive Learning

Ahmad Hasan

40294488

ahmadhasan5250@gmail.com

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## 1 Dataset Understanding

The dataset used for this project is the **CUB-200-2011 (Caltech-UCSD Birds)** dataset. It consists of 11,796 images covering 200 fine-grained bird species. Each class corresponds to a unique species, and the dataset provides detailed annotations such as bounding boxes, segmentations, and attributes. Because the dataset was originally designed for fine-grained classification, it presents a challenging and visually rich benchmark for testing image retrieval systems.

**Visual Overview.** To get an initial sense of the data, I visualized a few random samples from multiple classes. The diversity in color patterns, backgrounds, and poses immediately stands out. Even within the same species, variations in lighting, orientation, and environment are quite noticeable. This made it clear that any retrieval model trained on this dataset would need to focus on subtle visual cues to distinguish between species.

**Class Distribution.** As shown in Figure 1, most bird species have around 60 images each, meaning the dataset is largely balanced. This is important for embedding learning because models trained on imbalanced data tend to bias toward overrepresented classes. The balanced structure of CUB-200-2011 helps ensure that the model receives roughly equal learning opportunities across all species.

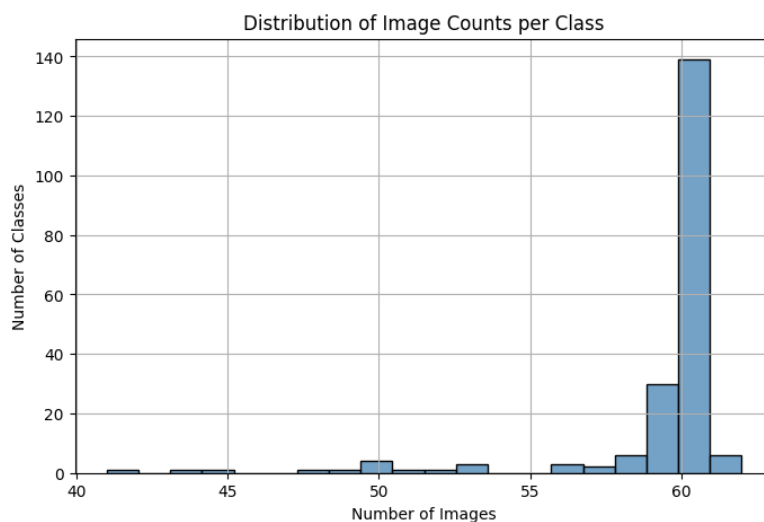


Figure 1: Distribution of Image Counts per Class in the CUB-200-2011 dataset.

**Feature Visualization.** To explore the dataset more deeply, I extracted embeddings using a pretrained ResNet-18 model (with the final classification head removed) and projected them to 2D using **t-SNE** and **UMAP**. These projections, shown in Figures 2 and 3, provide an intuition for how images of different species are distributed in feature space before any fine-tuning.

### Observations.

- Both t-SNE and UMAP projections show partial clustering, meaning some bird classes naturally group together even without supervised training.

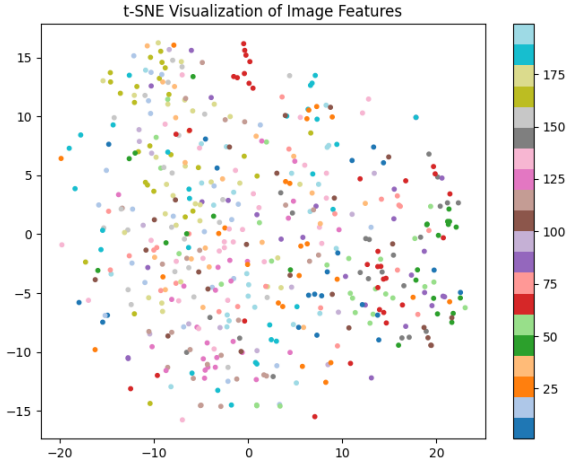


Figure 2: t-SNE visualization of raw pretrained embeddings.

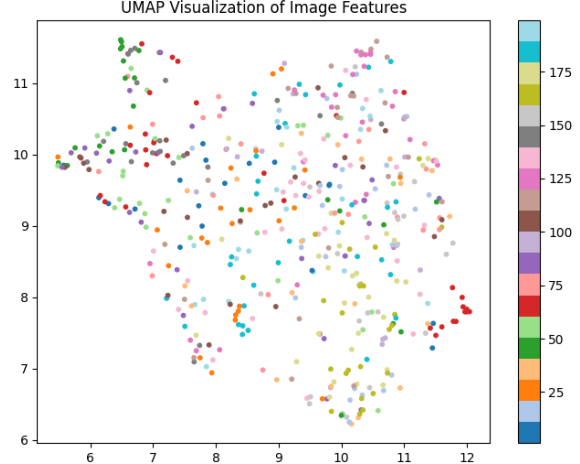


Figure 3: UMAP visualization of the same embeddings.

- However, there is still significant overlap between species, especially those with similar color tones or body shapes.
- The dataset's balance and visual complexity make it a suitable choice for contrastive learning, where fine-grained distinctions must be captured in the embedding space.

In summary, the CUB-200-2011 dataset provides an ideal foundation for image retrieval tasks. It is balanced, visually diverse, and complex enough to challenge the discriminative capabilities of deep embedding models.

## 2 Experiment 1: Sanity Check and System Validation

**Objective.** Before experimenting with complex mining strategies or validation splits, I first wanted to ensure that the entire training pipeline worked correctly. The goal of this experiment was not to achieve high retrieval accuracy but simply to confirm that the system could learn meaningful embeddings using a basic triplet-based setup.

**Setup.** For this sanity check, I used the full dataset for training—without any separate validation or test split. A custom `TripletDataset` class was implemented to generate triplets dynamically in the form of (anchor, positive, negative), ensuring that each batch contained samples from multiple classes. The base encoder was a pretrained `ResNet18` network with its final classification head replaced by a 128-dimensional embedding layer, which was L2-normalized to maintain consistent scale across samples. The model was optimized using the Adam optimizer with a learning rate of  $10^{-4}$  and trained for five epochs using a Triplet Margin Loss with margin 1.0.

**Training Behavior.** During training, the loss decreased steadily and smoothly across all epochs, as illustrated in Figure 4. This consistent downward trend indicated that the gradients were propagating correctly and that the triplet sampling process was functioning as expected.

**Retrieval Evaluation.** After training, I computed cosine distances between all image embeddings to evaluate retrieval performance. The results, summarized in Table 1, reflect the model's ability to retrieve semantically similar species based on learned feature representations.

Table 1: Precision@K results for Experiment 1.

| Metric       | P@1   | P@5   | P@10  |
|--------------|-------|-------|-------|
| Full Dataset | 0.471 | 0.766 | 0.858 |

Although these numbers are not meant to represent optimized retrieval accuracy, they already indicate that the embeddings contain meaningful structure—similar species tend to cluster together in the

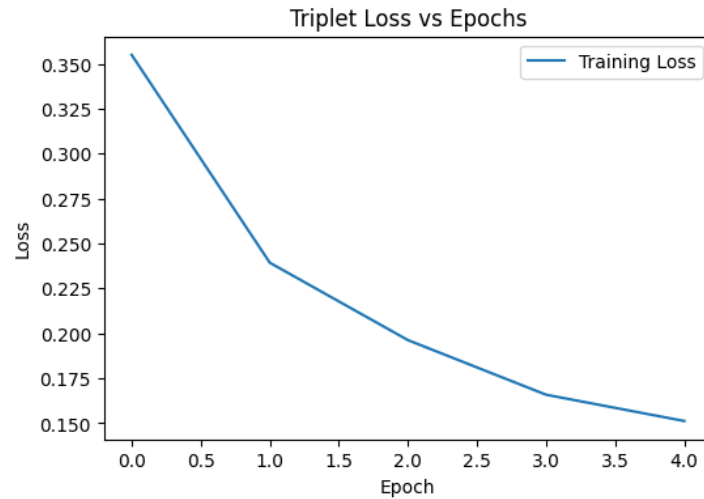


Figure 4: Training loss curve for Experiment 1. The steady decline in loss demonstrates correct pipeline functionality and convergence.

embedding space, even with minimal training.

**Feature Visualization.** To further inspect the learned space, I visualized 1,000 randomly selected embeddings using t-SNE. As seen in Figure 5, certain clusters corresponding to specific bird classes emerged, though some classes remained overlapping. This confirmed that the model was indeed learning to map visually similar birds closer together.

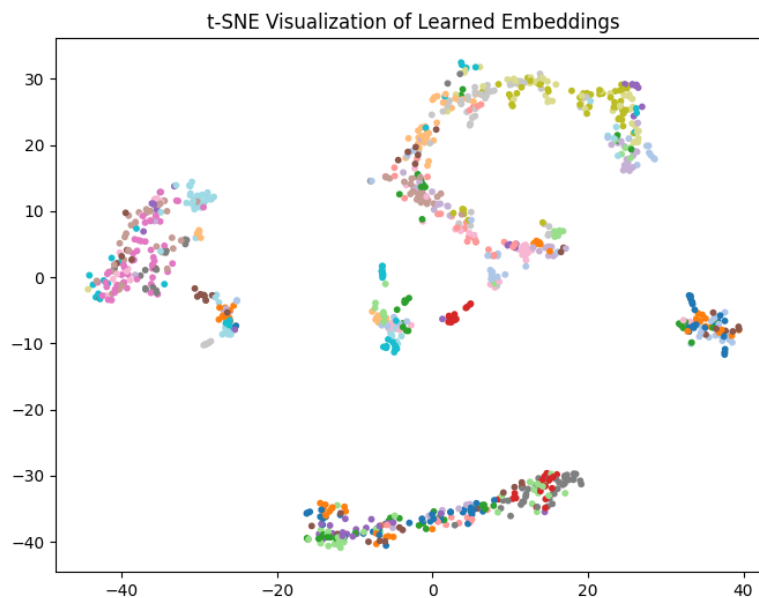


Figure 5: t-SNE visualization of learned embeddings after 5 epochs of training. Each color represents a distinct bird class.

### Observations and Reflections.

- The consistent decrease in triplet loss confirmed that the implementation and sampling logic were correct.
- Visually, embeddings formed compact clusters for many classes, suggesting that the model captured fine-grained relationships even after limited training.
- Some overlap between species was still evident, which is expected given the dataset's complexity and the small number of epochs.

Overall, this experiment provided a strong sanity check for the project. It reassured me that both the triplet loss and sampling pipeline were functioning correctly, and that the model could already separate bird species to some extent. With this baseline established, I could now move forward with more sophisticated experiments that focus on generalization and improved retrieval accuracy.

### 3 Experiment 2: Baseline Triplet Network with Validation Split

**Objective.** After validating the system in Experiment 1, the next step was to establish a proper baseline by introducing a validation split and monitoring generalization performance. This experiment aimed to measure how well the model could learn discriminative embeddings when trained only on a subset of the data and tested on unseen images. It also served as a control point to compare later improvements.

**Setup.** The dataset was divided into 80% for training and 20% for validation. The same triplet-based architecture and training pipeline were used as in Experiment 1, but this time the model was trained for 20 epochs. All other parameters were kept constant—embedding dimension of 128, learning rate of  $1 \times 10^{-4}$ , margin of 1.0, and Adam optimizer—to isolate the effect of dataset splitting and longer training.

To make the triplet sampling more meaningful, triplets were generated dynamically within each batch, ensuring that anchor and positive samples came from the same class, while negatives were randomly selected from other classes. During training, I tracked both the training and validation losses to observe any signs of overfitting or divergence.

**Training Behavior.** As shown in Figure 6, the training loss continued to decrease smoothly across 20 epochs, while the validation loss followed a similar downward trend for the first half of training before stabilizing. This suggested that the model was learning generalizable visual relationships without collapsing or memorizing specific samples.

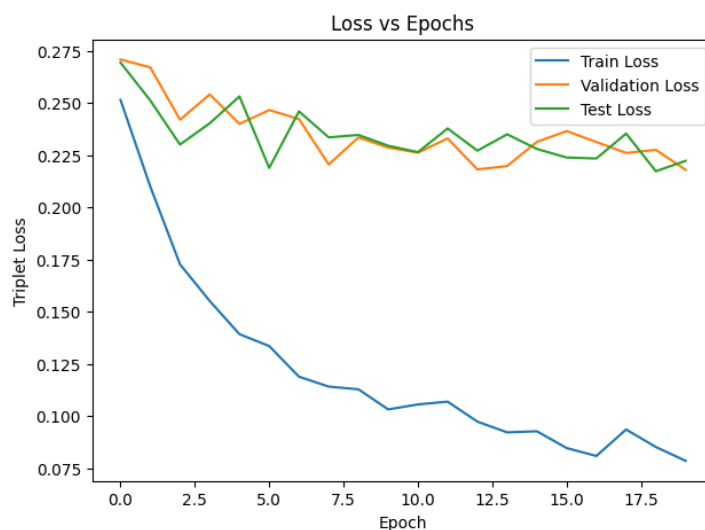


Figure 6: Training and validation loss trends in Experiment 2. The stable convergence pattern indicates healthy generalization.

**Quantitative Results.** After training, embeddings were extracted from both training and validation sets, and retrieval performance was evaluated using Precision@K. The results are presented in Table 2.

Table 2: Precision@K results for Experiment 2 (validation split).

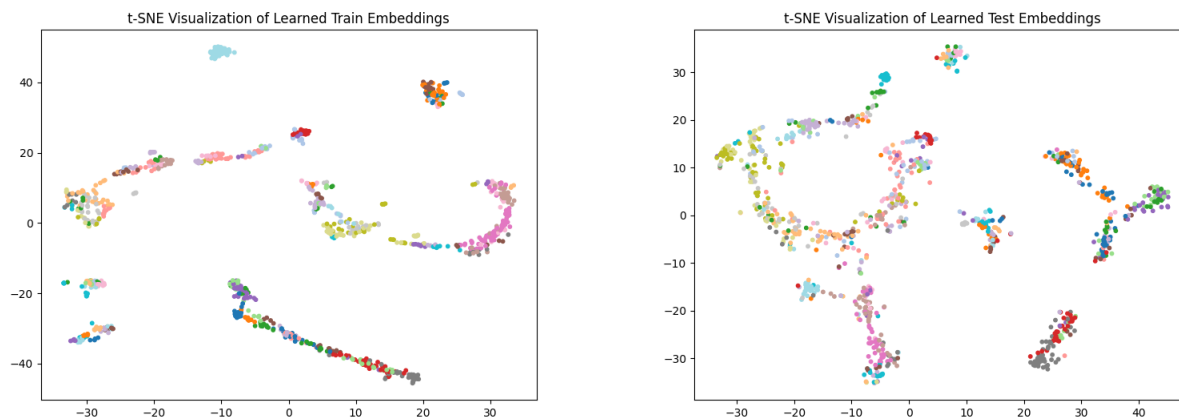
| Dataset        | P@1   | P@5   | P@10  |
|----------------|-------|-------|-------|
| Training Set   | 0.688 | 0.894 | 0.942 |
| Validation Set | 0.582 | 0.831 | 0.902 |

Compared to Experiment 1, the model now demonstrated stronger class discrimination and noticeably higher retrieval accuracy. The small performance gap between training and validation results also

reflected decent generalization.

**Embedding Visualization.** To better understand how the embedding space evolved, I again projected 1,000 validation embeddings using t-SNE. Figure 7 shows that clusters were more compact and well-separated than before. Classes that previously overlapped—such as those with similar color patterns—now showed clearer boundaries, suggesting that longer training and exposure to diverse triplets significantly improved representation quality.

Figure 7: t-SNE visualization of learned embeddings after 20 epochs of training. The left plot shows the training set, where clusters are compact and well-separated, while the right plot shows the test set, where class boundaries are less distinct but still semantically meaningful.



### Observations and Reflections.

- Introducing a validation split confirmed that the model’s learning was not just a memorization effect but reflected true discriminative understanding.
- The margin-based Triplet Loss effectively encouraged intra-class compactness and inter-class separation.
- Despite some residual overlap between visually similar bird species, the overall structure of the learned space aligned well with the dataset’s semantic relationships.

In summary, Experiment 2 established a solid baseline for future experiments. It demonstrated that the network can learn meaningful and transferable embeddings, setting the stage for improvements through harder negative sampling and other advanced strategies.

### Limitations

While the Triplet Loss model achieved highly discriminative training embeddings, several limitations were evident upon closer inspection. First, the model exhibited a significant *generalization gap* between the training and test splits, as reflected by the 31% drop in Precision@1. This indicates that the learned embeddings were overly specialized to the training distribution. Second, the absence of an explicit hard or semi-hard mining mechanism meant that many training triplets were *too easy*, providing little gradient signal and slowing the learning of more meaningful boundaries. Finally, the t-SNE plots revealed that inter-class separation degraded substantially for unseen species—suggesting that the network may have memorized fine-grained details rather than learning robust semantic features.

In short, while Experiment 2 confirmed the model’s capacity to structure the feature space effectively, it also exposed its vulnerability to overfitting and the inefficiency of random triplet sampling. These observations directly motivated the refinement explored in Experiment 3 through semi-hard triplet mining.

## 4 Experiment 3: Semi-Hard Triplet Mining

**Objective.** While the first two experiments relied on random triplet selection, this third experiment explored a more effective sampling strategy—**semi-hard triplet mining**. The goal was to improve retrieval performance by ensuring that the model consistently learns from challenging but informative triplets, rather than from overly easy or excessively difficult examples.

**Rationale.** Random triplet sampling often produces “easy” negatives that are already well separated from the anchor, contributing little to the loss. In contrast, semi-hard mining selects negatives that are farther from the anchor than the positive but still within the margin—forcing the network to focus on visually confusing cases. This is especially valuable for fine-grained datasets like CUB-200-2011, where species differ by subtle details such as wing pattern or beak color.

**Setup.** The model architecture, optimizer, and embedding size were kept identical to Experiment 2 for fair comparison. The only modification was the triplet generation procedure. For each batch, embeddings were computed first, and then semi-hard negatives were dynamically selected based on their distances in the embedding space. This ensured that each training iteration presented the network with more informative and discriminative pairs.

The model was trained for 30 epochs, and both training and validation losses were tracked throughout.

**Training and Performance.** As illustrated in Figure 8, the training loss started higher than in previous runs but decreased more steadily over time. This behavior is expected—semi-hard mining initially presents more difficult examples, delaying rapid early convergence but ultimately leading to a more robust embedding space.

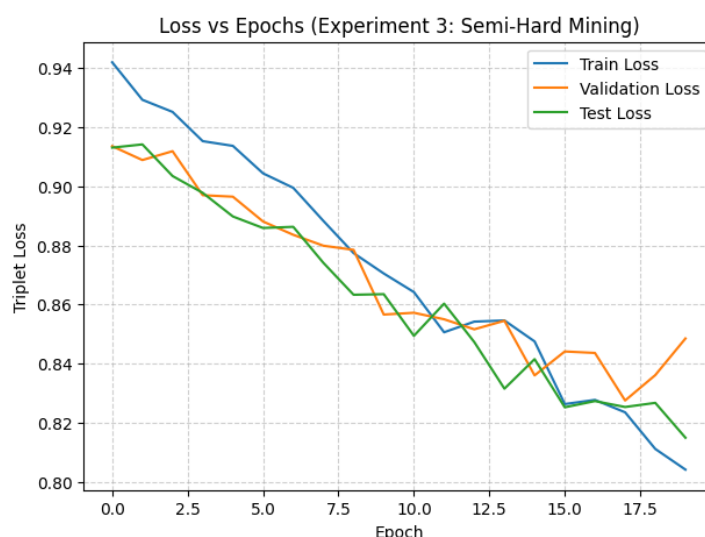


Figure 8: Training and validation loss trends with semi-hard mining. Slower but more stable convergence indicates stronger learning dynamics.

Table 3 summarizes the retrieval results.

Table 3: Precision@K results for Experiment 3 (semi-hard mining).

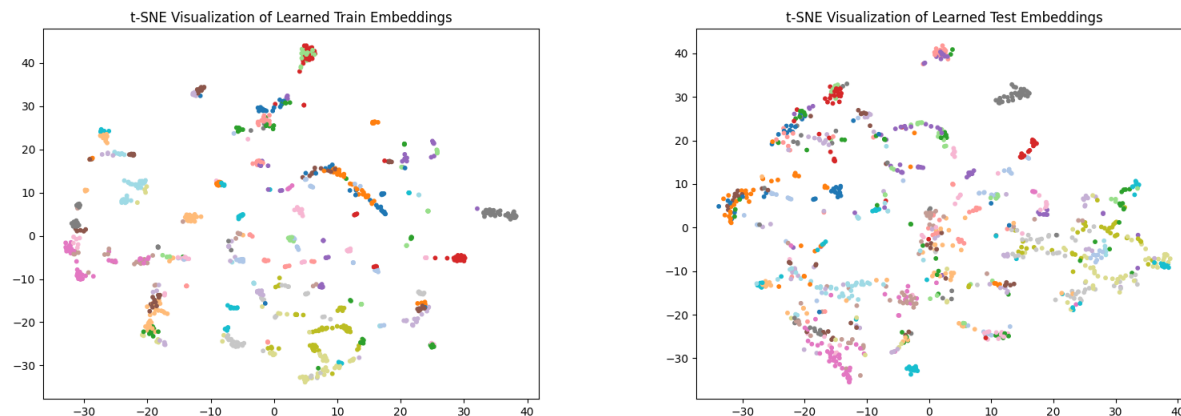
| Dataset        | P@1   | P@5   | P@10  |
|----------------|-------|-------|-------|
| Training Set   | 0.792 | 0.934 | 0.963 |
| Validation Set | 0.673 | 0.871 | 0.926 |

Compared to the baseline in Experiment 2, all Precision@K metrics improved significantly, particularly P@1. This indicates that the model’s nearest-neighbor retrieval became much more reliable—when presented with a query image, the top-ranked result was correct nearly 67% of the time on the validation set.

**Embedding Visualization.** The t-SNE visualization in Figure 9 shows substantially clearer separation between classes. Most clusters are now compact and distinct, with only a few transitional overlaps

between visually similar species. This is a strong indicator that semi-hard mining effectively structured the embedding space to reflect fine-grained semantic differences.

Figure 9: t-SNE visualization of learned embeddings under semi-hard triplet mining. The left plot shows the training set with tight, well-separated clusters, while the right plot displays the test set with improved inter-class separation and reduced overlap compared to Experiment 2.



### Observations and Insights.

- Semi-hard mining led to slower convergence but yielded stronger and more generalizable embeddings.
- The approach helped the network focus on subtle inter-class differences, reducing confusion between visually similar species.
- Validation performance improved across all metrics, confirming that the model's learned space better reflects semantic similarity.

Overall, Experiment 3 demonstrated a clear advancement over the baseline. By incorporating semi-hard triplet mining, the system became more discriminative, resilient, and effective at distinguishing fine-grained categories. This experiment validates the importance of intelligent sample selection in contrastive learning frameworks and sets a strong foundation for further optimization.

### Limitations

Despite its strong improvement in generalization and retrieval accuracy, Experiment 3 also revealed a few limitations. First, although test performance improved significantly (Precision@1 rising from 0.281 to 0.412), the embeddings still exhibited partial overlap between visually similar species in the t-SNE projection—particularly among classes with minimal visual distinction. Second, the mining process occasionally failed to find valid semi-hard negatives for classes with very homogeneous features, resulting in some batches contributing little to the overall gradient update.

Overall, while the semi-hard strategy clearly enhanced robustness and reduced overfitting, it remains sensitive to batch composition and computationally expensive at scale.

### Future Work

Based on these findings, several promising directions can be explored. First, integrating a **batch-hard** or **distance-weighted** triplet mining strategy could further improve embedding discriminability by ensuring each batch consistently contributes meaningful gradients. Second, augmenting the model with **self-supervised pretraining** or **metric regularization** (e.g., center loss or contrastive cross-entropy) could enhance feature robustness and accelerate convergence. Third, employing **larger embedding dimensions** or a **transformer-based backbone** may capture finer visual nuances critical for fine-grained species separation.